Solution Overview: Credit Risk Portfolio Monitor

I built a web dashboard that helps a bank's risk team see which loans are likely to go bad, why, and what to do about it. It covers 14,685 loans from 10,000 customers ($512.6M lent out), of which 16.1% have defaulted. It also has two AI helpers: a chat assistant that answers questions about the data, and an alert list of unusual accounts.

1. What the dashboard shows and how an analyst uses it

- **Portfolio view – "Where is the risk?"** Charts of risk by loan type, income, age, region and job type, plus filters to zoom into any group and see the exact loans inside it.
- **Account view – "Why is this loan risky?"** The chance this loan defaults, its risk level (Low / Medium / High / Very High), the main reasons behind the score, and its monthly payment history.
- **Anomaly review – "What needs attention now?"** A list of accounts behaving unusually, each with the reason. The analyst marks each one as a real problem or a false alarm.
- **Ask the Portfolio** – a chat box on every screen for questions that are not on a chart.

2. What I found in the data

- **Mortgages are the biggest risk.** They are only 1 in 10 loans but hold 58% of the money, and 1 in 4 of them defaults. *Action:* review these first.
- **Low income is the strongest warning sign.** Customers in the lowest income group default 2.7 times more often than the highest (25.5% vs 9.5%). Job type matters mostly at low income: low-income contract workers default at 27.5%, high-income ones at only 10.7%. *Action:* judge job type together with income.
- **The first missed payment is the early alarm.** Default rises from 13% (no missed payments) to 18%, 25% and 34% after one, two and three misses. *Action:* contact the customer after the first miss.
- **Data quality:** the data was mostly clean. A few values were missing or written differently (e.g. "M" instead of "Male"); I fixed them without deleting any row. Collections records were kept out of the model because they only exist after a customer has already defaulted, which would be cheating.

3. The two AI features

Ask the Portfolio (chat). The analyst types a question in plain English, e.g. "Which loan type defaults the most?". An AI model (Google Gemini) works out which calculation is needed, the app runs that calculation on the real data, and the AI writes a short answer with the source. **The AI never makes up numbers**: every number in the answer is checked against the data before it is shown. It can also score a new applicant, and it politely refuses questions the data cannot answer, like future interest rates.

Anomaly detection (alerts). The app compares each account with its own past behaviour (e.g. payments suddenly much later than usual) and with similar customers (same loan type and income level). 938 accounts are flagged, and they default at 25% compared with 16% overall, so the alerts point at real risk.

	Ask the Portfolio	Anomaly detection
Trigger	An analyst asks a question.	Runs every time the data is refreshed.
False positive	It answers a question it should have refused, or answers a slightly different question.	A good customer is flagged, e.g. one late payment because of a bank holiday.
False negative	It refuses a question the data could have answered.	A risky customer is missed because the change was too slow or happened at another bank.
Human check	Any number that cannot be verified is marked in the chat, and every step is logged.	The analyst confirms or dismisses each alert with a note; every decision is recorded.

4. The default prediction model

I trained two models to predict which loans will default: a simple, easy-to-explain one (logistic regression) and a more complex one (XGBoost). I tested them on customers they had never seen.

Model	AUC	Gini	KS	In simple words
Logistic regression (chosen)	0.68	0.35	0.27	Picks the riskier of two customers correctly about 68% of the time
XGBoost	0.67	0.34	0.27	Same level, but harder to explain

- **AUC** shows how well the model ranks risky above safe customers (0.5 = guessing, 1.0 = perfect). **Gini** and **KS** measure the same thing on other scales. These scores are typical for credit data where default is hard to predict.
- I chose the simpler model because it is just as accurate and every decision can be explained to a customer or a regulator. The model's risk levels match reality: loans marked "Very High" really default 40% of the time, "Low" only 5%.
- After going live, I would re-check accuracy every month and retrain it if the customer mix or accuracy changes.

5. Assumptions, and what I would need from the bank

- **Assumptions:** the money at risk is the loan's current unpaid balance; the default labels in the data are correct; the data is synthetic, so the patterns are realistic but not from a real bank.
- **From IT and data teams:** direct access to the bank's live data with an automatic daily refresh, a database to store analyst decisions, secure login for staff, an approved company AI service instead of the free one, and the real outcomes of loans so the model can be monitored over time.